Ethan Hartman <ehartma6@charlotte.edu>

[EXTERNAL] QORVO_RADIO2 Scope of Work Updates

2 messages

Nathan Whitt <Nathan.Whitt@qorvo.com> Fri, Sep 5, 2025 at 3:46 PM
To: Ethan Hartman <ehartma6@charlotte.edu>
Cc: Daisy Vazquez <Daisy.Vazquez@qorvo.com>, Russ Kegley <David.Kegley@qorvo.com>, Jed Sander <Jed.Sander@qorvo.com>, Nan BouSaba <nbousaba@charlotte.edu>, Fareena Saqib <fsaqib@charlotte.edu>

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Hey Ethan,

From our discussion this morning, I mentioned that I would provide some deliverable and requirement details to put into the statement of work. I've also attached the brief presentation we were viewing.

This list of deliverables and requirements should be pretty exhaustive in terms of our requirements for the project. If the scope seems too large, please let us know. Please review the items with your team, update the statement of work accordingly, and send the statement of work back to us when you're ready for us to review it before or at our next meeting. Also, please let us know if you have any questions and we'll answer ASAP.

The list is below, and also attached as a .txt file if that's easier for you to view:

Main project goals (minimum/required) - IoT Application Demonstration (Use QPG6200 in application devices, with some existing gateway as a link to web-based control)

Deliverables:

- Hardware/physical:

- Demo application hardware, including:

- User interface and application controls (e.g. buttons/sensors, controller, phone/tablet etc.)

- IoT application PCB assemblies (PCB assemblies designed by team that contain radio and sensor or transducer)

- IoT gateway (bought device, something like EW840E5N... possibly)

- Software:

- Demo application software (e.g. webpage code, mobile application code, etc.)

- Embedded system software (e.g. "microcontroller" program(s))

- IoT gateway configuration (if applicable)

- Documentation:

- Bill of Materials

- Schematics

- Layout and design files

- Software documentation

- IoT gateway documentation if config is non-standard

- User guide/application note(s)

- Written feedback for Qorvo dev kit and software (no specific format)

Quantifiable Requirements:

- Deliverables must operate using at least two (2) separate IoT protocols (e.g. Zigbee, Thread over Matter, Bluetooth, etc.)

- Deliverables must include three (3) separate application use cases (e.g. remote microphone, lighting, remote sensing, etc.

- Deliverables must implement a total of at least three (3) peripherals from the QPG6200 device (e.g. PDM, PWM, I2S, ADC's, UART, USB)

- Deliverable remote IoT devices must be coin cell battery powered.

- Deliverable remote IoT device assemblies must use a custom PCB design specific for intended applications.

- (Preferred) Custom PCB design should be a common PCB assembly. *** (If different designs are better, that is OK. To review during concept design.)

- Deliverable remote IoT devices must be able to connect to an internet (not necessarily the Internet, but that is OK).

- Deliverable remote IoT devices must be able to communicate bi-directionally with a webpage* user interface. (*Or mobile application)

- Deliverable remote IoT devices must be able to operate in a mesh topology (e.g. Bluetooth Mesh, mesh nodes with Matter over Thread).

- Deliverables must implement a battery life calculation for remote devices to be displayed on the control interface.

Qualitative requirements (verification of these requirements can be noted as qualitative in SoW, and may even be subjective to personal opinion.):

- Demonstration deliverables should be highly interactive and involve creative use cases/application devices.

- Demonstration deliverables should be visually engaging and easy to understand.

- Demonstration deliverables should demonstrate connectivity interoperability using IoT devices. (e.g. different protocols work in conjunction)

- Demonstration deliverables should include hands-on interaction (e.g. employ sight/touch/sound/voice, etc.).

- Demonstration deliverables should include enclosures.

Specifications:

- Remote devices must have battery life of > 1 week

- Size:

- Remote devices must occupy a space less than 3"x3"x3"

- Weight:

- Remote devices must weigh less than 0.5 lb

Standards/Regulations:

- Follow FCC guidelines, specifically for ISM band 2.4GHz - 2.5GHz

- IoT standards:

- Bluetooth

- Zigbee

- Matter (over Thread)

- BLE

Secondary (stretch) goals - Demonstration of QPG6200 application boards using custom software defined radio (SDR) system as the IoT gateway

Deliverables:

- Hardware/physical:

- New front end receive hardware for existing SDR - MAY only require a small fix to existing hardware.

- New front end transmit hardware for existing SDR

- Software:

- Any software implementation with some SDR as a gateway (doesn't have to be existing hardware)

- Documentation:

- BOM, schematics, layout/design files for SDR

- User guide

- Software documentation

Quantifiable requirements:

- SDR hardware must operate on frequencies between 2.4GHz and 2.5GHz.

- SDR system must fit inside the existing case.

- SDR should operate as a minimal gateway (it should be able to send and receive data between the control interface and remote devices in SOME way).

Qualitative requirements:

- SDR implementation should be easy to use by future UNCC students.

Specifications:

- No particular specifications

Standards/Regulations:

- Same as above/main project

Thank you,

Nathan Whitt
Mobile Test Engineer | Qorvo
8220 Piedmont Triad Pkwy
Greensboro, NC 27409
(Building M, 1st Floor, 1N03-SW, MS11)
E Nathan.Whitt@qorvo.com
M +1 828.493.4935

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2 attachments

QORVO_RADIO2_scope_of_work_details.txt
5K

Senior Design Kickoff.pdf
415K

Ethan Hartman <ehartma6@charlotte.edu> Fri, Sep 5, 2025 at 4:20 PM
To: Abdul Bah <abah2@uncc.edu>, Luke Matthews <lmatth23@uncc.edu>, Alberto Hernandez <aherna86@uncc.edu>, Kenneth Le <kle26@uncc.edu>

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